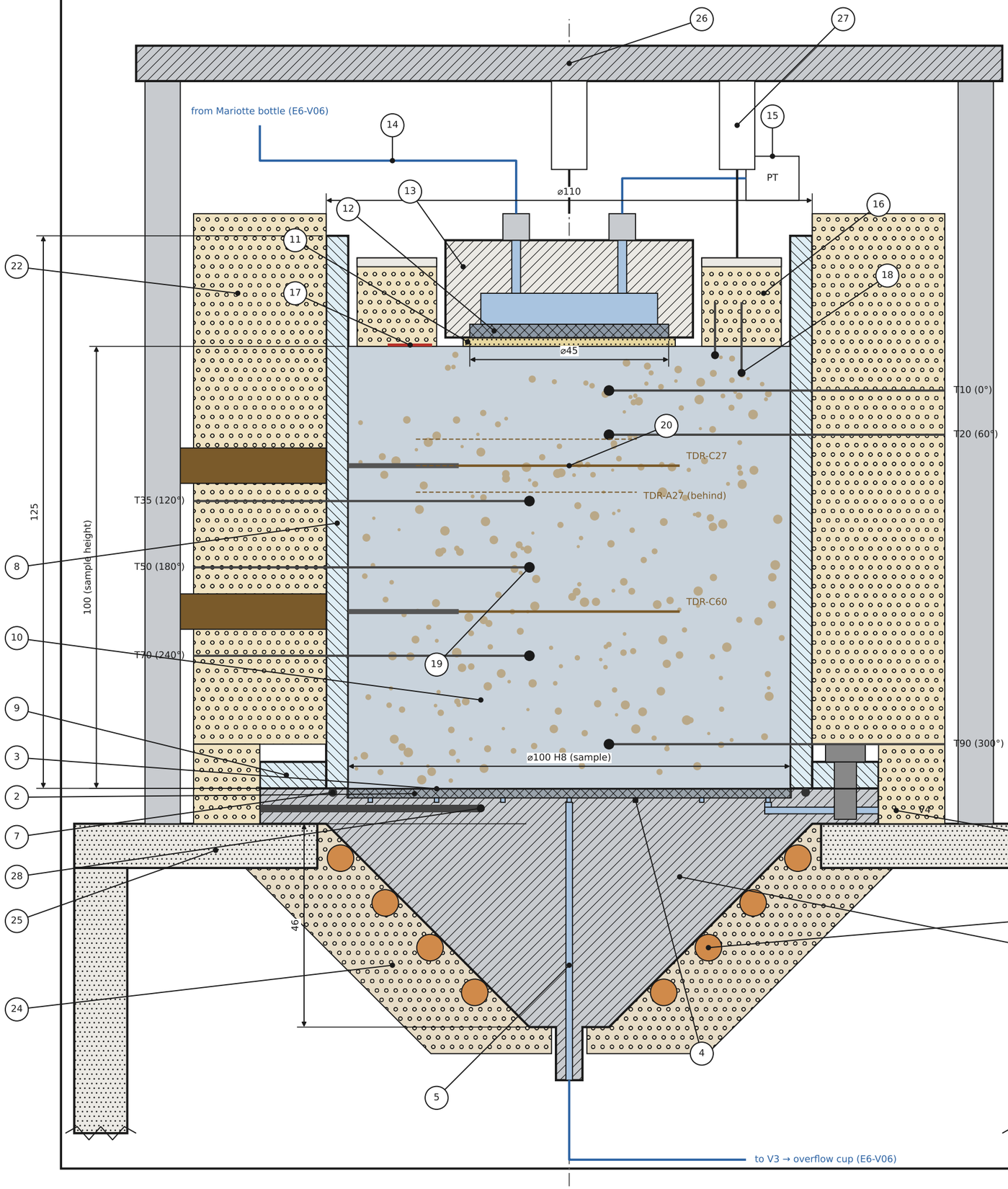


Parts list

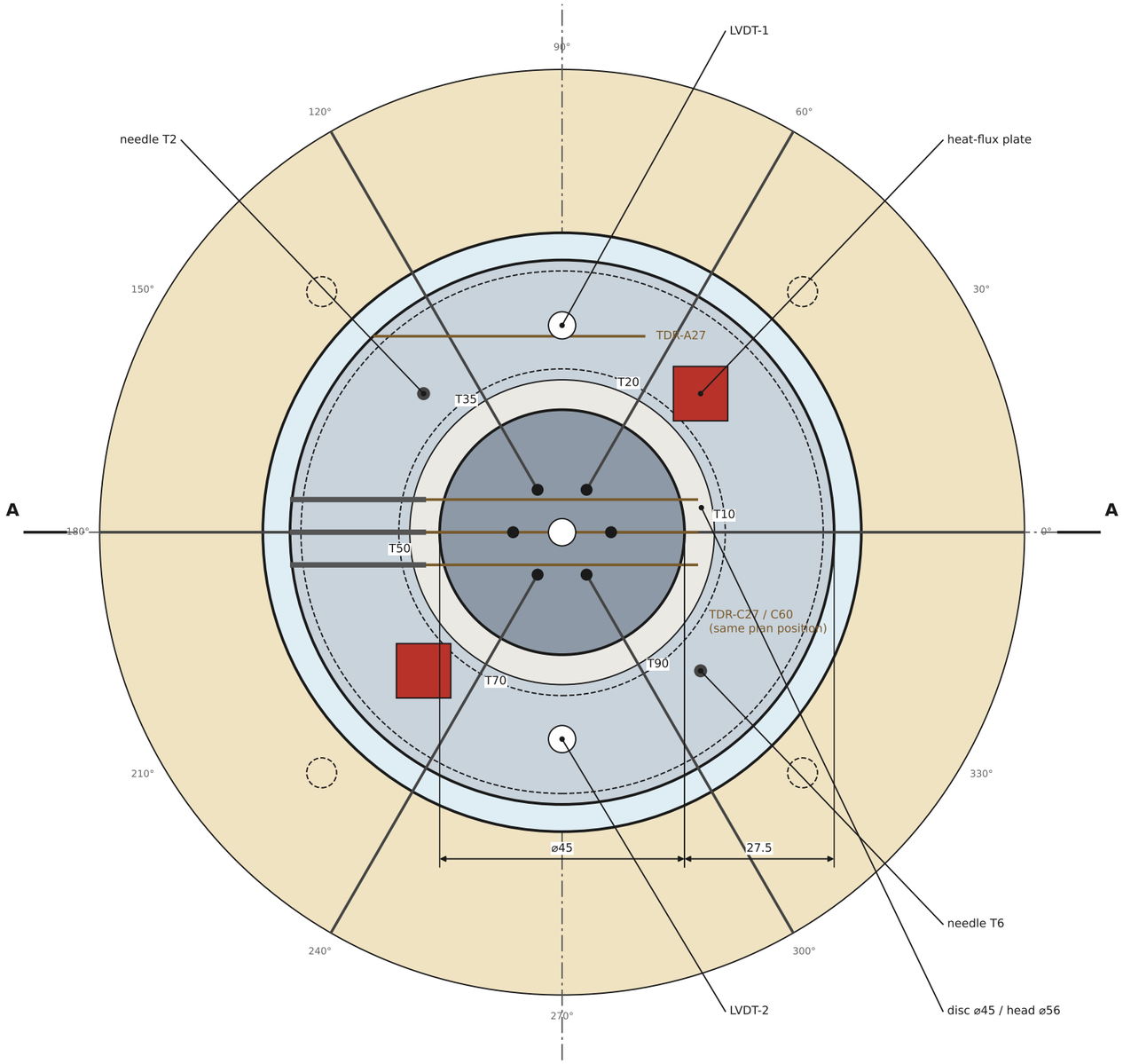
1. cone outlet body, Al 6082, solid, 45° (E6-V04)
2. perforated support plate AISI 316, t 2, holes ø1.5 / pitch 4
3. nylon membrane 20 µm, ø104, clamped by tube end face
4. drainage grooves 1×1: 12 radial + 3 rings (r 15/30/45)
5. axial bore ø1.5 → nozzle ø6 → PTFE 1/16" → valve V3
6. air-purge port from ring groove r 45, valve V4
7. O-ring ø107×2 NBR, face seal
8. PMMA tube ø110/100×125, inner wall sand-coated (epoxy)
9. PMMA flange ø140×6, bonded; 4× M5 into cone rim
10. soil sample ø100×100, packed in 10 mm lifts
11. contact sand 0.1–0.3 mm, t 2, ø48, placed dry
12. sintered disc ø45×3, stainless (as Mini Disk / AMDI)
13. AMDI head body, PETG – low-profile variant, 2 ports
14. supply line from Mariotte bottle on balance, FEP 4/2
15. diff. pressure sensor at disc ±2.5 kPa (+ air purge)
16. floating insulation ring XPS ø96/60×20, PETG frame
17. heat-flux sensor 10×10 mm, 2× at r 36 (45°, 225°)
18. thermistor needles in ring, 2 and 6 mm (135°, 315°)
19. wall thermistor ø1.6, tip at r 9, 6× helix at 60°
20. centre TDR C27, C60: 3 rods ø1.6×75, 25 mm insulated
21. annulus TDR A27: chord y 36, 3 rods ø1.6×50, stacked
22. side insulation Kaiflex t 30
23. Cu coil ø6, 4 turns on cone → circulating bath
24. cone insulation, PU foam shell t 20
25. stand plate POM t 10, 3 legs
26. LVDT bridge, Al 20×8 on posts from stand plate
27. LVDT ±10 mm: 0 – head, 1/2 – ring (90°, 270°)
28. thermistor in cone, 4.5 mm below membrane

z = 0: top face of membrane = bottom of sample.
Sensor depths refer to the initial surface z = 100.
Item 21 lies behind the section plane (dashed).

axisymmetric section; off-plane items revolved into the section plane (angles: E6-V02)

Column assembly – section A–A

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E6 – infiltration into frozen soil	date 2026-10-20
scale	**1 : 1**	drawing no.	**E6-V01**	sheet 1 / 12



Sensor schedule

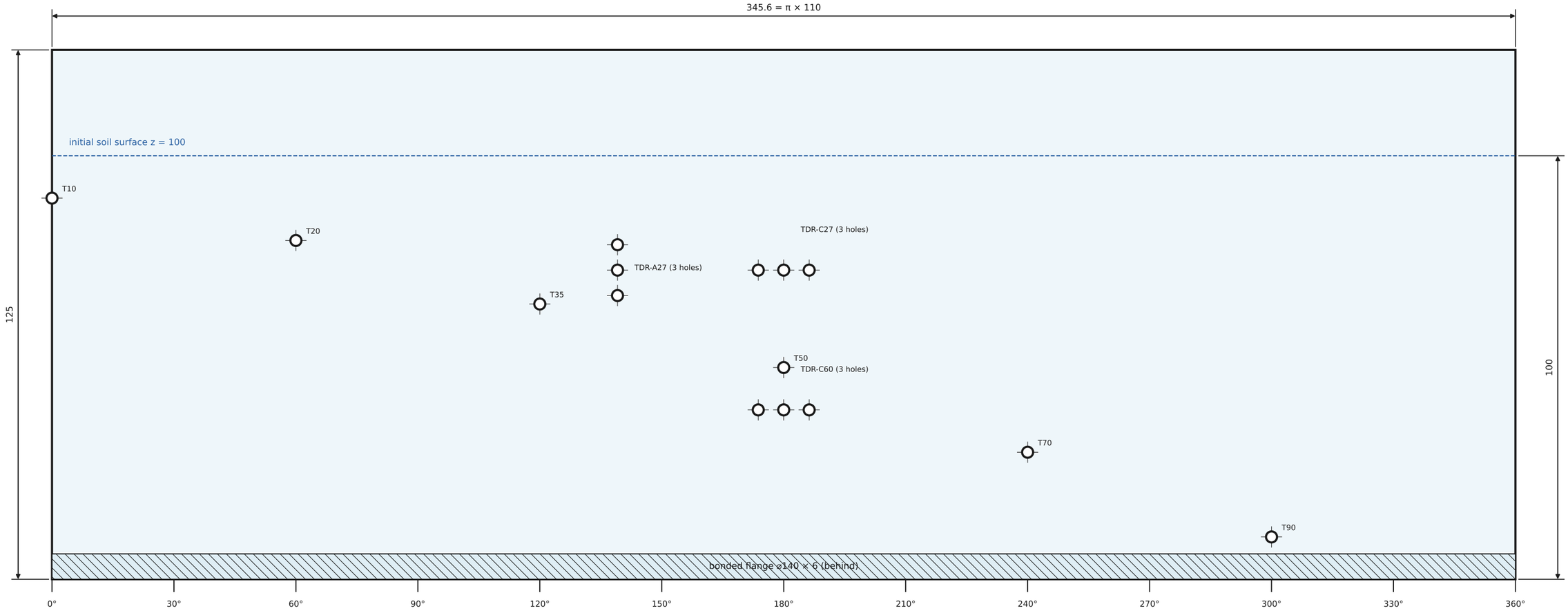
ID	type	angle °	r / y	depth	z
T10	thermistor, wall	0	9	10	90
T20	thermistor, wall	60	9	20	80
T35	thermistor, wall	120	9	35	65
T50	thermistor, wall	180	9	50	50
T70	thermistor, wall	240	9	70	30
T90	thermistor, wall	300	9	90	10
T2r	thermistor needle, ring	135	36	2	floats
T6r	thermistor needle, ring	315	36	6	floats
Tdisc	thermistor, disc/sand	180	7	0	floats
Tcone	thermistor, cone	180	20	–	-4.5
TDR-C27	TDR 3×ø1.6×75, 25 ins.	180	y=0	27	73
TDR-A27	TDR 3×ø1.6×50	chord	y=36	27	73
TDR-C60	TDR 3×ø1.6×75, 25 ins.	180	y=0	60	40
HF1	heat-flux 10×10	45	36	0	floats
HF2	heat-flux 10×10	225	36	0	floats
LVDT-0	LVDT ±10 mm	–	0	–	top
LVDT-1	LVDT ±10 mm	90	38	–	top
LVDT-2	LVDT ±10 mm	270	38	–	top

depth = mm below the initial surface; z = height above the membrane.
Ring sensors float with the surface: they record heave / thaw settlement via LVDT-1/2 and keep a fixed depth below the moving surface.

angle 0° = +x (right half of section A–A), counter-clockwise seen from above

Sensor layout – plan view

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E6 – infiltration into frozen soil	date 2026-10-20
scale	**1 : 1**	drawing no.	**E6-V02**	sheet 2 / 12



Hole table

ID	angle °	arc mm	z	hole	direction	purpose
T10	0.0	0.0	90	⌀2.0	radial	thermistor ⌀1.6 + gland
T20	60.0	57.6	80	⌀2.0	radial	thermistor ⌀1.6 + gland
T35	120.0	115.2	65	⌀2.0	radial	thermistor ⌀1.6 + gland
T50	180.0	172.8	50	⌀2.0	radial	thermistor ⌀1.6 + gland
T70	240.0	230.4	30	⌀2.0	radial	thermistor ⌀1.6 + gland
T90	300.0	288.0	10	⌀2.0	radial	thermistor ⌀1.6 + gland
TDR-C27.1	186.3	178.8	73	⌀2.0	parallel to x	TDR rod ⌀1.6
TDR-C27.2	180.0	172.8	73	⌀2.0	parallel to x	TDR rod ⌀1.6
TDR-C27.3	173.7	166.8	73	⌀2.0	parallel to x	TDR rod ⌀1.6
TDR-A27.1	139.1	133.5	67	⌀2.0	parallel to x (41° to wall)	TDR rod ⌀1.6
TDR-A27.2	139.1	133.5	73	⌀2.0	parallel to x (41° to wall)	TDR rod ⌀1.6
TDR-A27.3	139.1	133.5	79	⌀2.0	parallel to x (41° to wall)	TDR rod ⌀1.6
TDR-C60.1	186.3	178.8	40	⌀2.0	parallel to x	TDR rod ⌀1.6
TDR-C60.2	180.0	172.8	40	⌀2.0	parallel to x	TDR rod ⌀1.6
TDR-C60.3	173.7	166.8	40	⌀2.0	parallel to x	TDR rod ⌀1.6

Notes

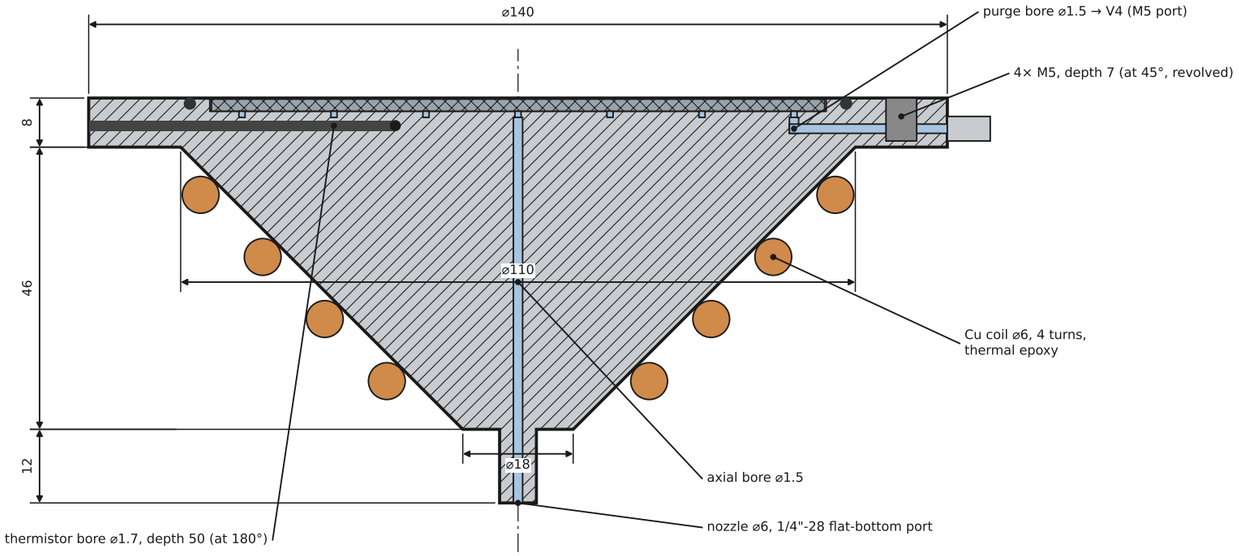
1 Arc length measured on ⌀110 from 0°; z from the tube end face (= membrane top).
2 Drill TDR holes on CNC with a fixture: rods are parallel to the x axis, not radial.
3 Seal glands with silicone (low-temperature grade); check at −10 °C before use.
4 Inner wall: thin epoxy + 0.3–0.5 mm sand coat against wall flow; mask holes.
5 Holes in the insulation jacket follow the same pattern.

PMMA tube ⌀110/100 × 125, seen from outside; angle increases to the right

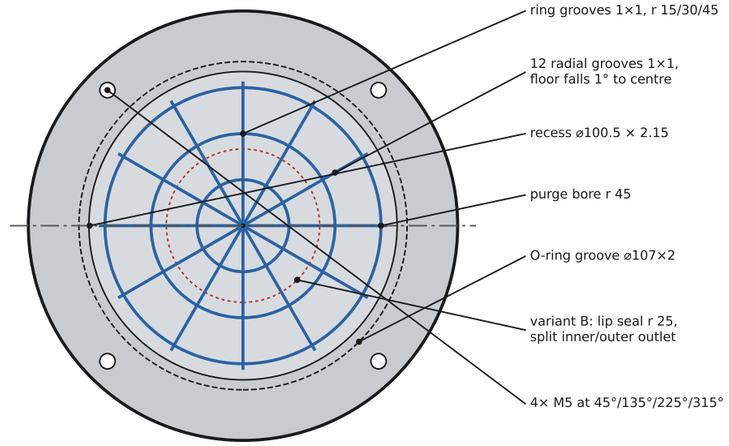
Column tube – developed outer surface

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E6 – infiltration into frozen soil	date 2026-10-20
scale	**1 : 1**	drawing no.	**E6-V03**	sheet 3 / 12

SECTION B–B 1 : 1

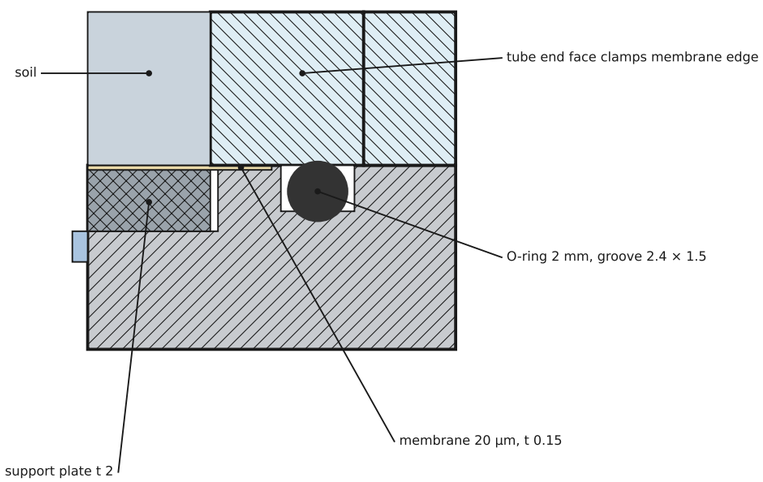


VIEW C – top face (membrane removed) 1 : 2



Why a solid cone
The external funnel keeps the drip point on the axis, but the inside is solid, so the drainage volume is grooves 1.1 mL + bore 0.1 mL (< 1 fraction) instead of the 120 cm³ air space of a hollow funnel. The membrane lets the bottom be held at a set suction h_b = 0 … −30 cm (overflow cup height, E6-V06); h_b = 0 reproduces a seepage face. The Cu coil keeps the cone at −10 °C while freezing and at +0.3 °C during outflow.

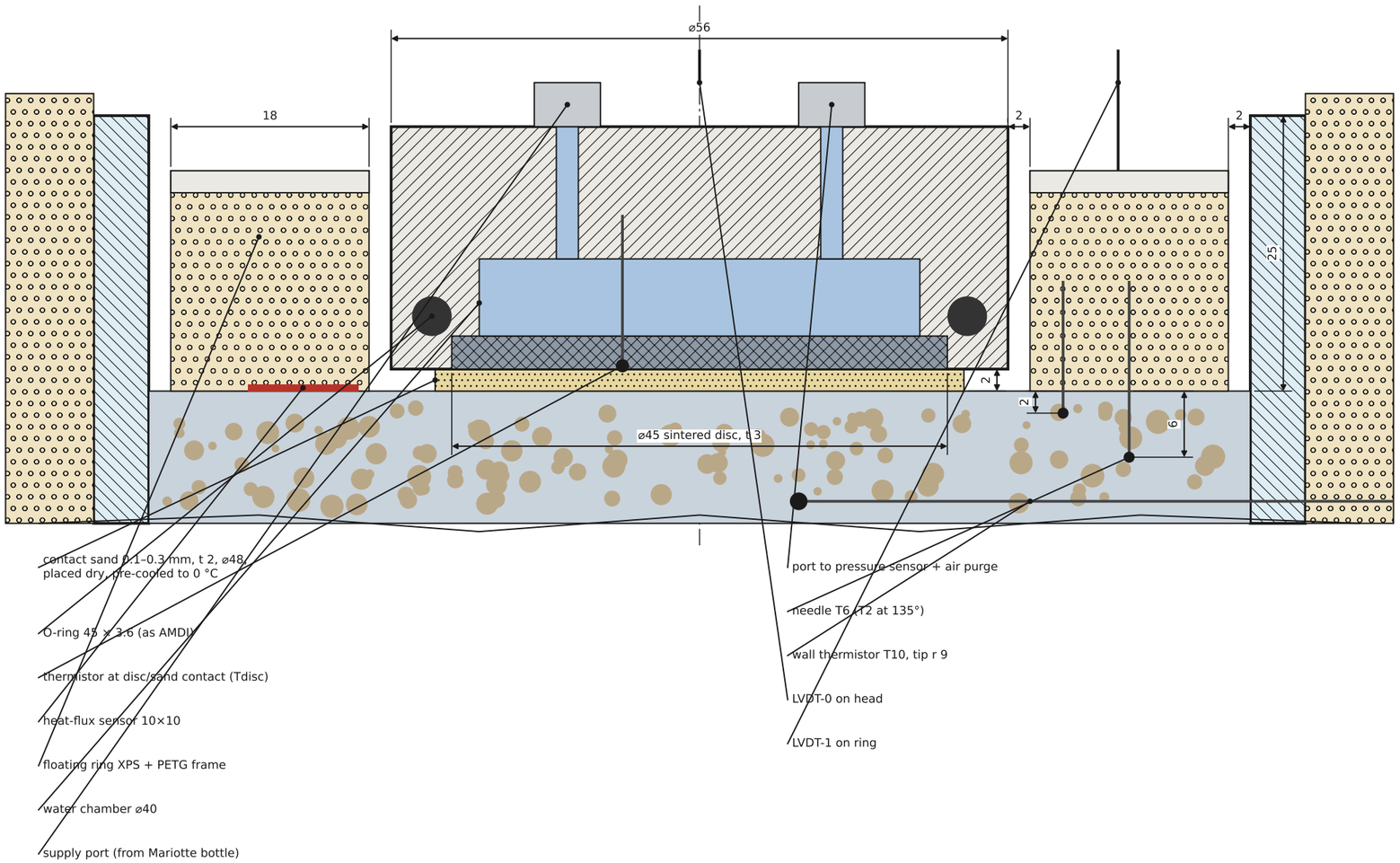
DETAIL D – membrane clamp 5 : 1



solid Al cone: external 45° funnel, internal drainage volume ≈ 1.2 mL

Cone outlet with suction membrane

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E6 – infiltration into frozen soil	date 2026-10-20
scale	**1 : 1 / 5 : 1**	drawing no.	**E6-V04**	sheet 4 / 12



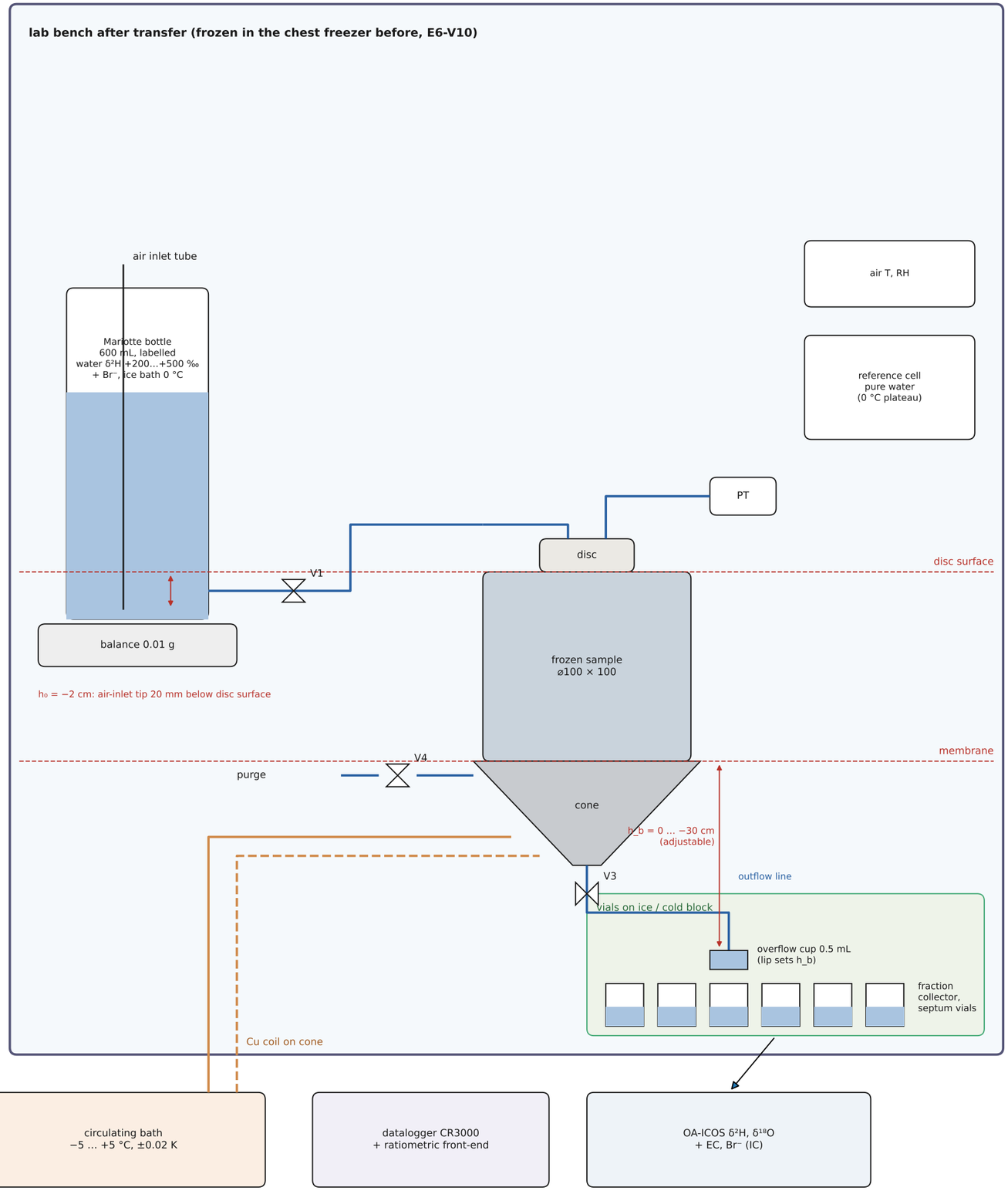
Head changes against the AMDI repository

- Keep the printed lower body, disc seat and O-ring; drop the on-board Mariotte tube.
- Two top ports: supply from the external Mariotte bottle on a balance, and a port to the differential pressure sensor that also serves as air purge.
- Head mass full ≈ 60 g (≈ 0.4 kPa on the disc) instead of ≈ 300 g (≈ 2 kPa).
- Gaps of 2 mm let head and ring float freely with frost heave / thaw settlement.
- Supply water and head stay at +0.5 °C; log Tdisc to detect freezing in the sand.

AMDI disc (Kubát et al. 2025, github.com/kubatjanf/AMDI), low-profile head

Disc–soil interface detail

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E6 – infiltration into frozen soil	date 2026-10-20
scale	**2 : 1**	drawing no.	**E6-V05**	sheet 5 / 12



Valves and boundary conditions

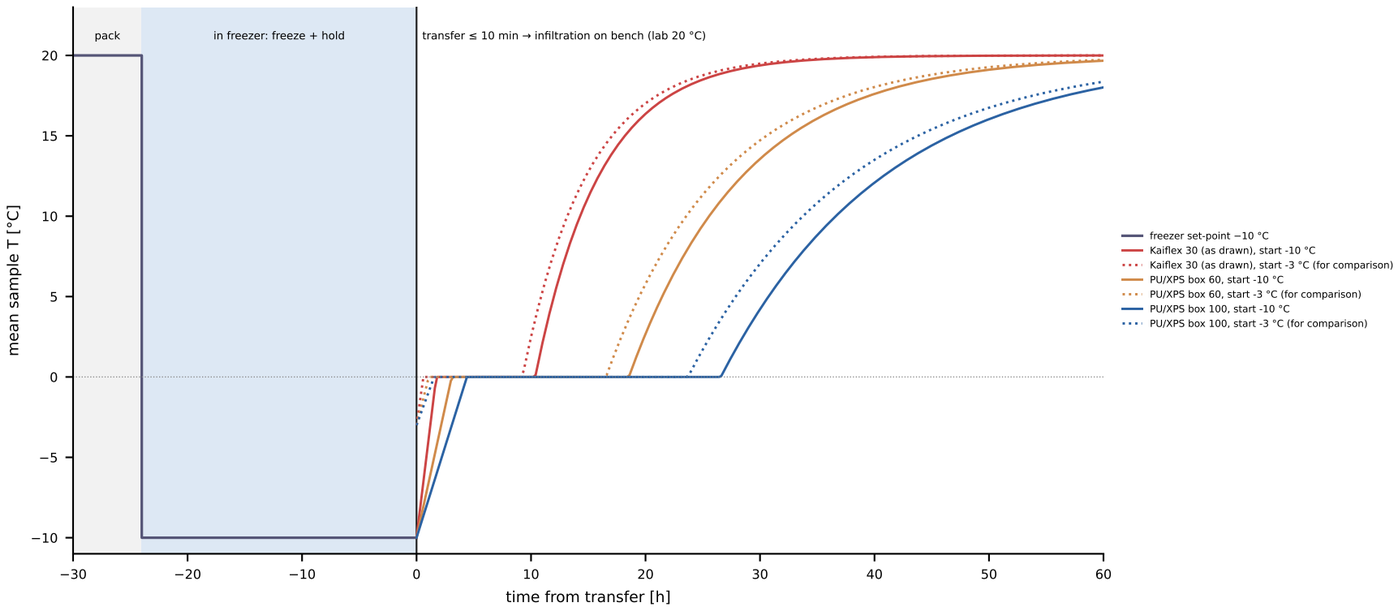
item	setting	note
V1	supply on/off	closed until infiltration start
V3	outflow	closed while freezing (stops cryosuction from the outlet line), opened at start
V4	air purge	flush grooves before freezing
top	$h_0 = -2$ cm	set by Mariotte air inlet; PT logs real h_0
bottom	h_b = 0…−30 cm	set by overflow-cup lip; h_b = 0 → seepage face (original design)
sides	≈ adiabatic	insulation box 60–100 mm; HF2 checks
cone	T_b	−10 °C freezing, +0.3 °C during outflow

```
Balances
water:   in (balance) − out (fractions) = ΔS; ΔS checked by TDR (liquid)
         + gravimetric sectioning at the end (total = liquid + ice)
tracer:  δ²H label, Br⁻; δ¹⁸O left unlabelled → indicator of ice–water
         fractionation and of pre-event water
heat:    T profile + 2 heat-flux sensors + cone bath power
```

not to scale except the marked elevations

Hydraulic and thermal scheme

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E6 – infiltration into frozen soil	date 2026-10-20
scale	–	drawing no.	**E6-V06**	sheet 6 / 12



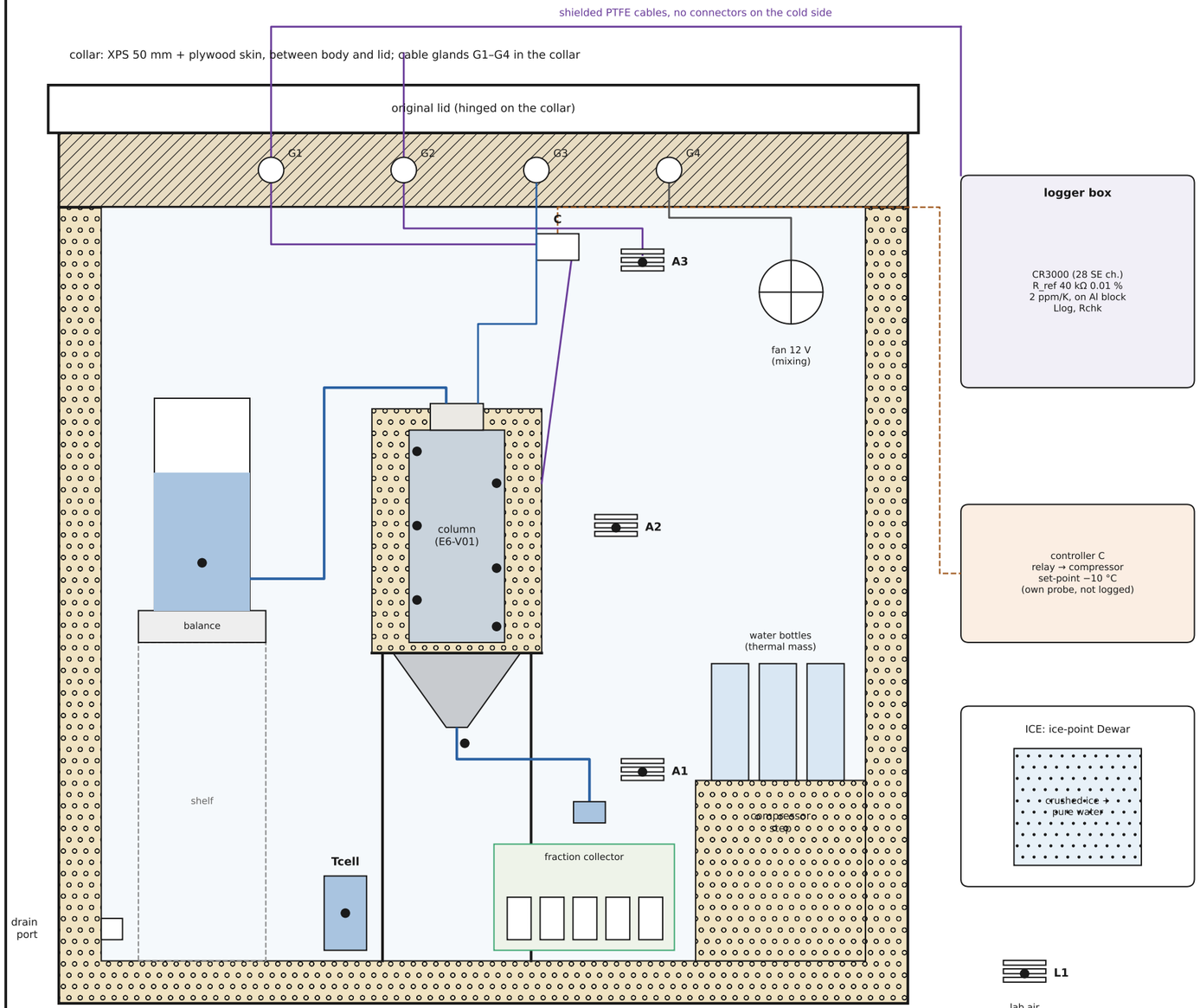
The 0 °C plateau is the time with ice in the pores. Breakthrough of the label takes 44–111 h, so every run covers infiltration into frozen soil, its thawing, and flow through thawed soil.

step	where	action	go / no-go
1	lab	pack Ø100×100 in 10 mm lifts to θ ≈ 0.30 with background water; mount holder on cone, all probes in, V3 closed, flush V4	probes read, TDR ok
2	freezer	freezer −10 °C, sides + cone insulated → freezing from the top through the Al plate; logger running through the collar	front passes T90
3	freezer	hold at −10 °C ≥ 24 h (controller: constant −10 °C, not the ±10 °C cycle program)	all T within ±0.05 K
4	prep	ice-water bath for Mariotte bottle and contact sand; collector rack on ice; insulation box open on the bench next to the freezer	Tsup ≤ 0.5 °C
5	transfer	lift column + cone + stand as one unit, cables stay connected; close box; write transfer start / end time	≤ 10 min
6	bench	dry pre-cooled sand + AMDI; open V1 and V3; collect 2–5 mL fractions in septum vials on ice	Tdisc logged
7	end	≥ 1.5 PV out or 120 h; close V1, V3; section in 10 mm slices: gravimetric θ, δ²H, Br^-	balance ±2 %

times indicative; bench phase from a lumped heat balance (E6-V12, vypocty.md §8)

Run protocol – freeze in freezer, infiltrate on bench

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E6 – infiltration into frozen soil	date 2026-10-20
scale	–	drawing no.	**E6-V07**	sheet 7 / 12



Temperature channels

ID	location	in/out
T10	soil, wall probe, depth 10 mm, r 9	in
T20	soil, wall probe, depth 20 mm, r 9	in
T35	soil, wall probe, depth 35 mm, r 9	in
T50	soil, wall probe, depth 50 mm, r 9	in
T70	soil, wall probe, depth 70 mm, r 9	in
T90	soil, wall probe, depth 90 mm, r 9	in
T2r	soil, ring needle, 2 mm below surface	in
T6r	soil, ring needle, 6 mm below surface	in
Tdisc	disc / contact sand	in
Tcone	cone body under membrane	in
Tsup	Mariotte bottle water	in
Tout	outflow at nozzle	in
A1	freezer air, floor level, shielded	in
A2	freezer air, column mid-height	in
A3	freezer air, under lid	in
Tcell	reference cell, 50 mL pure water	in
L1	lab air at freezer, shielded	out
Llog	logger enclosure, at R_ref	out
ICE	ice-point Dewar (spare probe)	out
Rchk	fixed 40 kΩ resistor (no NTC)	out

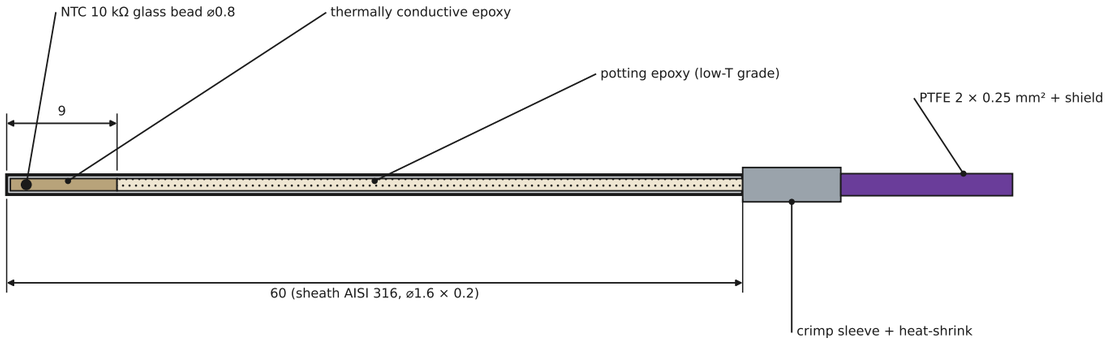
Rules
- Never drill the freezer walls (refrigerant lines inside); use the collar.
- One logger, one R_ref, one set of cables for calibration and runs.
- Logger stays outside at room temperature; log Llog next to R_ref.
- Excitation 0.5 V for ~50 ms per reading → self-heating ≈ 1.4 mK.
- Check ICE (0.000 °C) before and after each run; drift > 3 mK → recalibrate.
- Tcell shows the 0 °C freezing plateau of pure water during freezing: an in-situ check of the whole chain at the most important temperature.
- Fan + water bottles damp the compressor cycling (typ. ±2 K → < ±0.3 K at A2); the column insulation damps it further.

temperature channels inside and outside the freezer; not to scale

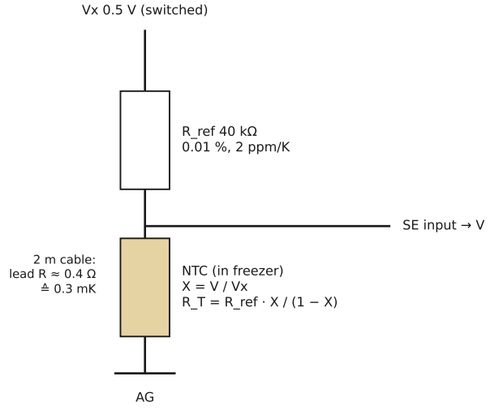
Thermometry – chest freezer layout

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E6 – infiltration into frozen soil	date 2026-10-20
scale	–	drawing no.	**E6-V08**	sheet 8 / 12

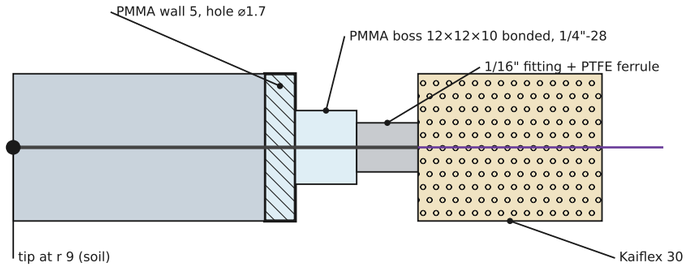
PROBE P1 (wall probe, tip at r 9) 2 : 1



HALF-BRIDGE (ratiometric)



INSTALLATION (section, 1 : 1)



Calibration procedure

1 Assemble probes with their final cables; connect to the run logger.
2 Stirred bath (ethanol–water), reference SPRT/Pt100 with certificate.
3 Points: −12, −10, −8, −5, −3, −1, 0 (ice point), +1, +5, +10, +20 °C;
   ≥ 20 min per point, 10 readings averaged, probes within 20 mm of ref.
4 Fit Steinhart–Hart per probe: $1/T = a + b \ln R + c (\ln R)^3$;
   accept if all residuals < 2 mK.
5 Ice-point check of every probe before and after each run.
6 Recalibrate every 6 months or when ICE drift > 3 mK.
7 Store coefficients with probe ID in the data repository.

Bridge sensitivity

T °C	R_T kΩ	X	0.01 K ≙ µV	self-heat mK
-10	59.0	0.5958	69	1.0
-5	44.5	0.5265	69	1.0
0	33.9	0.4587	66	1.0
5	26.1	0.3948	61	1.0
20	12.6	0.2389	42	0.8

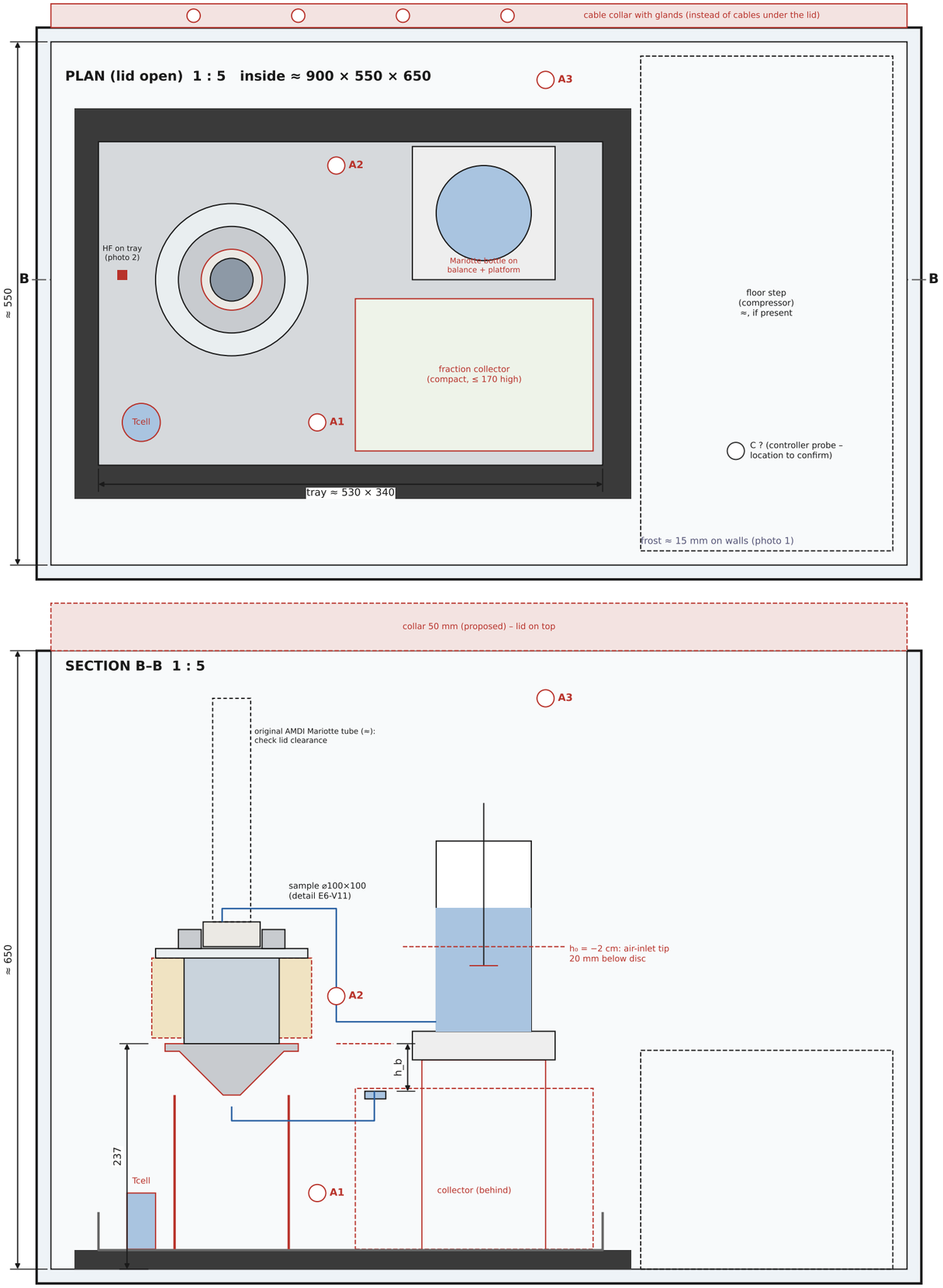
Uncertainty budget (mK)

component	u
reference thermometer (SPRT / Pt100, cert. U = 5 mK, k = 2)	2.5
bath inhomogeneity and stability	2.0
Steinhart–Hart fit residual	1.0
R_ref drift / tempco (2 ppm/K × 15 K, same logger as in calibration)	0.6
logger resolution and noise (24 bit, averaged)	0.3
self-heating difference bath vs. soil	1.0
sensor drift between calibrations (checked at ice point)	2.0
combined standard uncertainty u_c	4.1
expanded uncertainty U (k = 2)	8.2

probe ø1.6 = 1/16" so it seals in a standard HPLC compression fitting

Thermistor probe, circuit and calibration

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E6 – infiltration into frozen soil	date 2026-10-20
scale	**2 : 1 / 1 : 1**	drawing no.	**E6-V09**	sheet 9 / 12



Seen in the photos

1 Chest freezer, programmable controller (program Set A: 24 h +10 °C, then 192 h steps −10 / +10 °C, linear). Controller clock shows 29.02.2000.
2 Heavy frost on the walls; cables run under the lid seal.
3 Stainless tray ≈ 530 × 340 on a black insulation mat.
4 Sample in a clear printed PETG holder, ø100 inside, wide top flange. It stands directly on the tray (no outlet).
5 Al plate ≈ ø112 × 20 on top, centre hole ≈ ø26, recess ≈ ø60.
6 Two thermistor wires pushed vertically into the sand from the top.
7 Two heat-flux sensors taped with duct tape: on the Al plate, on the tray.
8 Black cable + blue tube enter the holder at the side (TDR? – confirm).

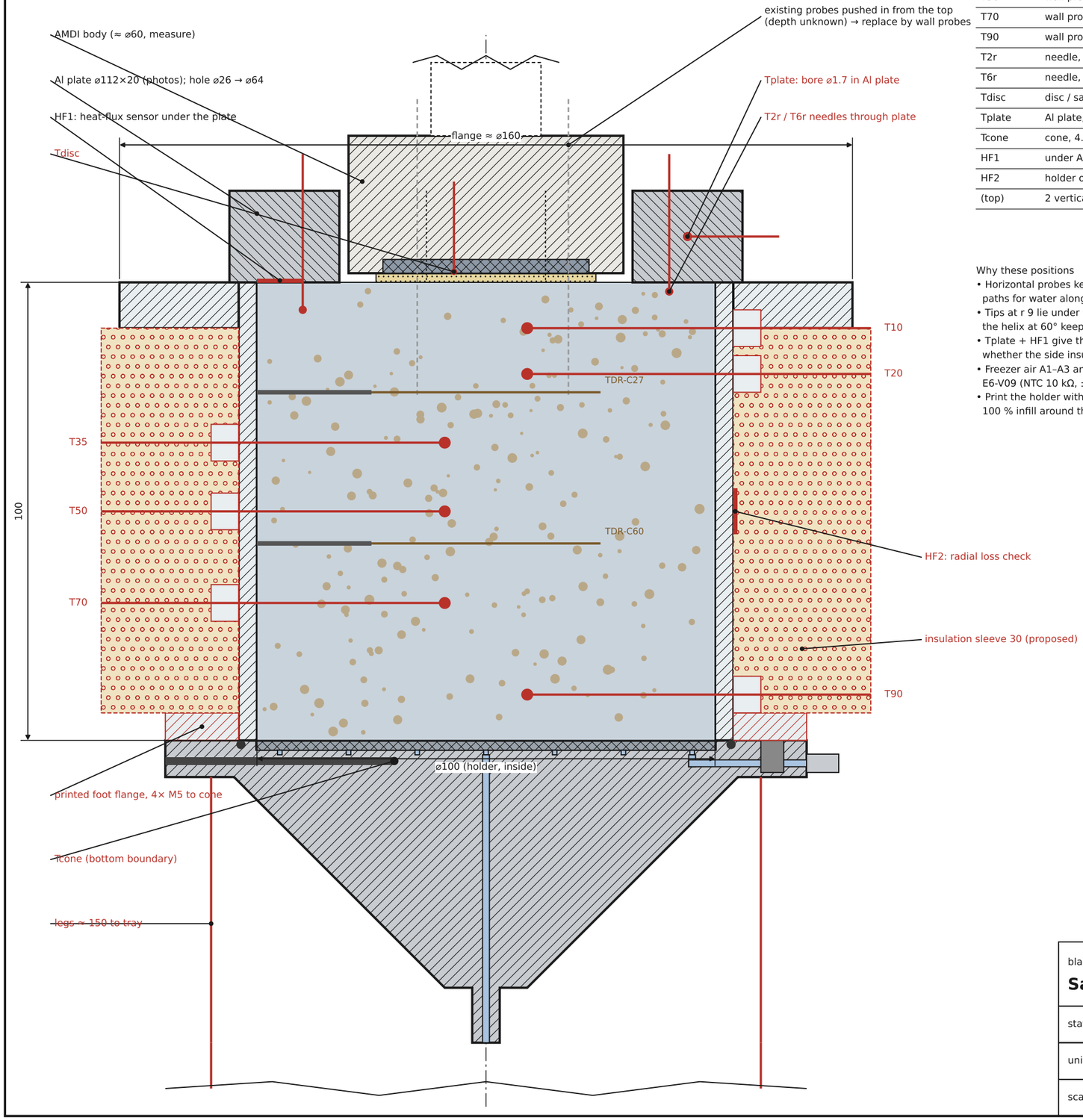
Changes (red)

A Raise the sample on the cone outlet + stand ≈ 150 mm; the collector sits beside it on the tray, overflow cup sets h_b (default −5 cm).
B Side insulation sleeve 30 mm around the holder (no radial freezing).
C Al plate: open the hole to ø64 so the AMDI disc reaches the soil and the plate stays as a cold annular top boundary.
D Cable collar with glands → lid seals, less frost, less sublimation.
E Defrost before each run; set the controller clock (time base for data).
F Mariotte bottle on balance + platform; air inlet 20 mm below the disc.
G Temperature sensors as in E6-V11 (+ A1–A3, Tcell, controller probe C).

dimensions marked ≈ are estimated from photos and must be measured

As-built chest freezer – realistic layout

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E6 – infiltration into frozen soil	date 2026-10-20
scale	**1 : 5**	drawing no.	**E6-V10**	sheet 10 / 12



Temperature positions

ID	position	angle	mounting	status
T10	wall probe, r 9, depth 10	0°	1/16" fitting in printed boss	new
T20	wall probe, r 9, depth 20	60°	1/16" fitting in printed boss	new
T35	wall probe, r 9, depth 35	120°	1/16" fitting in printed boss	new
T50	wall probe, r 9, depth 50	180°	1/16" fitting in printed boss	new
T70	wall probe, r 9, depth 70	240°	1/16" fitting in printed boss	new
T90	wall probe, r 9, depth 90	300°	1/16" fitting in printed boss	new
T2r	needle, r 40, depth 2	0°	through Al plate	new
T6r	needle, r 40, depth 6	180°	through Al plate	new
Tdisc	disc / sand contact, r 7	–	glued to AMDI disc rim	new
Tplate	Al plate, r 44, mid-thickness	0°	radial bore ⌀1.7 × 12	new
Tcone	cone, 4.5 below membrane	180°	radial bore	new
HF1	under Al plate, r 45	180°	move from top of plate	seen → moved
HF2	holder outer wall, z 50	0°	under insulation sleeve	seen → moved
(top)	2 vertical probes from top	–	remove	seen

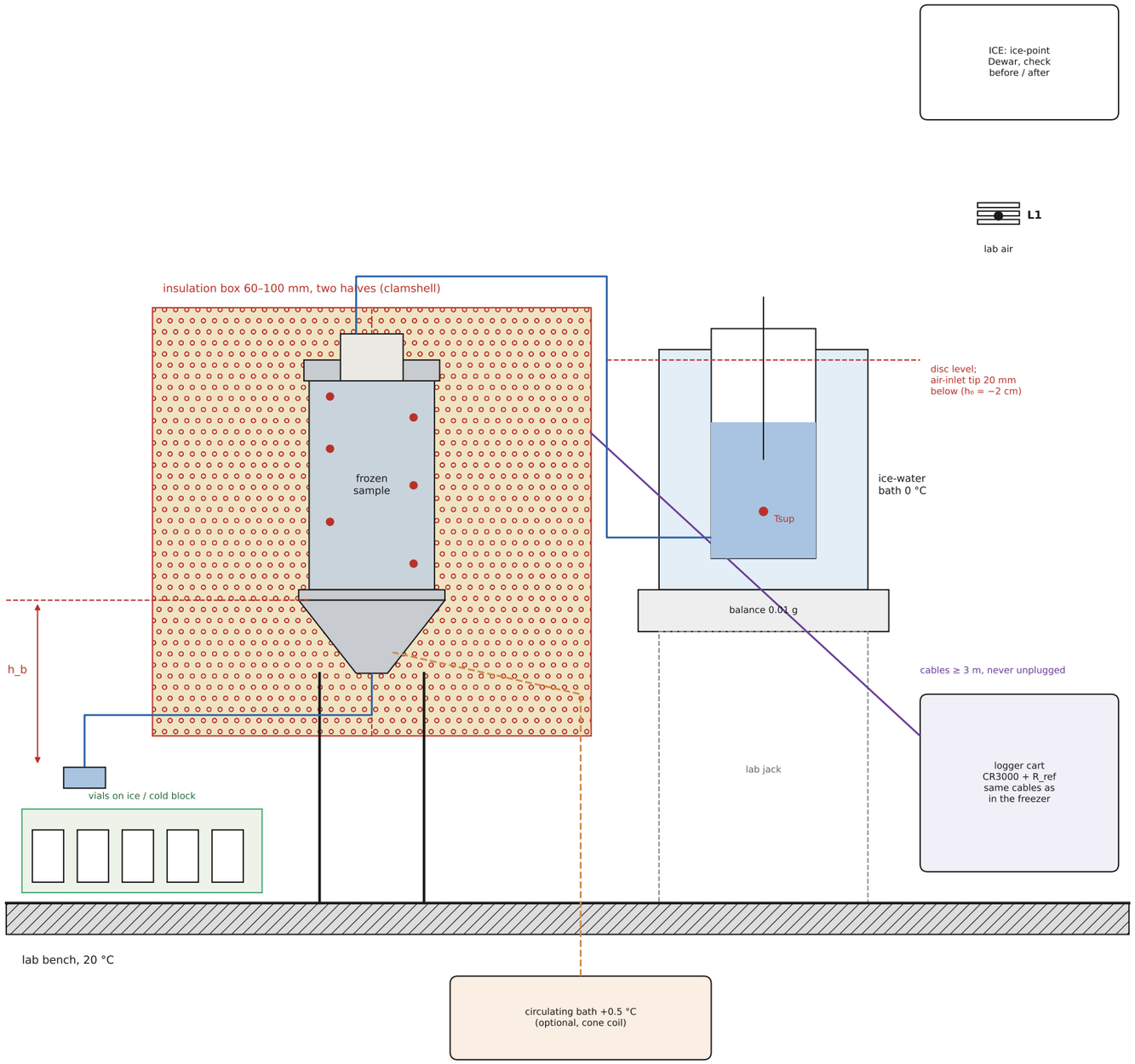
Why these positions
- Horizontal probes keep a defined depth and do not open vertical paths for water along the probe under the disc.
- Tips at r 9 lie under the disc (⌀45), where the infiltration front is; the helix at 60° keeps probe shadows from stacking.
- Tplate + HF1 give the top heat input, Tcone the bottom, HF2 tells whether the side insulation keeps the flow 1-D.
- Freezer air A1–A3 and Tcell: see E6-V10; electronics and calibration: E6-V09 (NTC 10 kΩ, ±0.01 °C after chain calibration).
- Print the holder with 12×12 mm bosses at the probe heights (PETG, 100 % infill around the bosses), thread 1/4"-28 after printing.

black = as in the photos, red = proposed; ⌀ of AMDI body and holder wall to be measured

Sample holder – temperature positions

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E6 – infiltration into frozen soil	date 2026-10-20
scale	**1 : 1**	drawing no.	**E6-V11**	sheet 11 / 12

Freezer → bench: lift column + cone + stand as one unit; the logger cart stands next to the freezer so the cables reach both places; temperature record stays continuous.



How long does it stay frozen on the bench?

insulation	start °C	UA W/K	to 0 °C h	ice gone h
Kaiflex 30 (as drawn)	-3	0.097	0.6	9
Kaiflex 30 (as drawn)	-10	0.097	1.7	9
PU/XPS box 60	-3	0.054	1.1	15
PU/XPS box 60	-10	0.054	3.1	15
PU/XPS box 100	-3	0.038	1.5	22
PU/XPS box 100	-10	0.038	4.4	22

Lumped estimate, lab 20 °C, conduction through the insulation only. Real times are shorter: uninsulated AMDI head, legs, cables, and warm infiltration water (84 J/mL at 20 °C vs 2 J/mL at 0.5 °C → use ice bath).

What it means

- Use the thickest practical box (≥ 60 mm) and start colder (−10 °C) (−10 °C chosen: ≈ 3 h before 0 °C with a 60 mm box; −3 °C would give ≈ 1 h).
- The run always continues into thawing soil: the probes record when and where the ice disappears (0 °C plateau ends).
- Supply water, sand and the AMDI disc at 0–0.5 °C, otherwise the disc melts its own footprint first.
- Bottom: a frozen cone blocks the outlet; the optional bath at +0.5 °C thaws the lowest mm so water can leave at h_b.
- Collect fractions on ice in septum vials: lab air would evaporate them.

infiltration outside the freezer; not to scale except marked heights

Bench set-up after transfer

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E6 – infiltration into frozen soil	date 2026-10-20
scale	–	drawing no.	**E6-V12**	sheet 12 / 12